Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/16/2024
PI Name:	Robling, Alexander Ph.D.		
Project Title:	Foundational In Vivo Experiments on Osteocyte Biology in Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Indianapolis	**State:**	IN
Zip Code:	46202-5126	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	02/28/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Klotz, Rebecca	**Contact Phone:**	650-604-1119
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 02/28/2025 per NSSC information (Ed., 3/17/24) NOTE: Extended to 11/01/2023 per F. Hernandez/ARC (Ed., 2/27/23) NOTE: Extended to 11/01/2022 per F. Hernandez/ARC and NSSC information (Ed., 10/15/21) NOTE: Extended to 11/01/2021 per NSSC information (Ed., 9/10/20) NOTE: Extended to 9/30/2020 per F. Hernandez/ARC (Ed., 5/1/19) NOTE: Extended to 5/31/2019 per NSSC information (Ed., 6/21/18) NOTE: Extension not yet approved; end date changed back to 5/31/2018 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: Extended to 3/31/2019 per F. Hernandez/ARC (Ed., 3/5/18)		
Key Personnel Changes/Previous PI:	No changes		

COI Name (Institution):	Chen, Jake Ph.D. (Indiana University) Feng, Jian Ph.D. (Texas A&M University System) Fraga, Mario Ph.D. (University of Oviedo) Sumner, Rick Ph.D. (Rush University Medical Center) Lee, Brendan M.D., Ph.D. (Baylor College of Medicine)
Grant/Contract No.:	NNX15AL13G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Bone loss due to the weightlessness of space is a major health concern for astronauts that spend significant amounts of time in space. The development of effective countermeasures to bone loss in space requires a detailed understanding of the biological alterations among cells that control skeletal homeostasis. Those alterations can take on many forms, including chemical modifications to the genome (epigenetics), changes in the expression of certain genes (RNA transcription) and certain small nucleic acid chains that regulate protein production (microRNA), and changes in the physical properties of bone matrix, among others. We have assembled a team of experts in bone biology to explore the changes in osteocyte biology and bone matrix properties that are associated with long-term space flight, with the goal of elucidating new mechanisms to target for maintaining bone health during space travel. To this end, we propose to study mice that have spent 5 weeks or 4 months aboard the International Space Station (ISS). We will look for changes in DNA, RNA, protein, using an “omics” approach to capture all of the changes induced by space flight. We will also study tissue level properties to investigate changes in bone mass, turnover dynamics, and material properties. Our experiments will provide insight into the limitations of ground-based models, and identify the biological processes for which they exhibit high fidelity to space flight changes. Moreover, we anticipate that new mechanisms fostering bone loss will emerge, which will provide new targets to counteract space travel-induced bone loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research impact and benefit to human health on Earth has not changed since the last report was submitted. The experiments we are conducting have impact on, and benefit to, the skeletal health not only of individuals that spend a considerable amount of time in space, but also to individuals on Earth that suffer from the bone-wasting effects of numerous musculoskeletal disorders, including neuromuscular paralysis, nonunited fractures, prolonged bedrest, and genetic diseases that compromise bone integrity. Our studies will likely also uncover new therapeutic treatment avenues for more common disorders such as postmenopausal osteoporosis or glucocorticoid-induced osteoporosis. We are using cutting edge technologies to identify alterations in gene expression, microRNA expression, protein modification, and DNA epigenetic regulation that occur as a result of microgravity. As such, we anticipate that several biomolecular target-based technologies will come from our studies.
Task Progress:	While waiting for a flight date, we have worked out many of the details of the flight conditions with the NASA Ames Research Center (ARC) Rodent Research (RR) Rodent Research Mission Science team. We are/were to the point where all analyses techniques are ready to go, as all of the parameters have been validated with ground-based experiments. We were given a flight date, but it was re-scheduled due to COVID-19 related issues. Further updates on an anticipated flight date and experiments were expected to be provided by NASA Space Biology program management based on flight opportunities and prioritization of the flight experiments. I don’t believe we will have the opportunity to fly the experiment, and the grant has gone through numerous no cost extensions (NCEs) as we wait. We are currently out of funding to do the space flight arm of the proposal, so a flight opportunity would require that we re-apply for a new grant.
Bibliography Type:	Description: (Last Updated:)